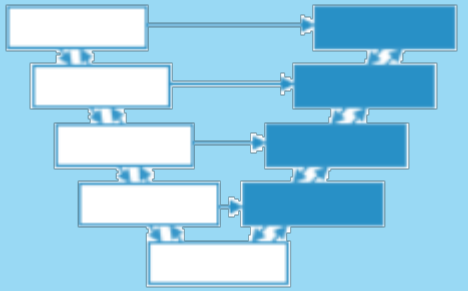
Collaboration, Quality, Test

Prof. Dr. Jóakim von Kistowski

Check-In

Recap:
What do we need to consider when testing in **iterative and incremental software development models**?

V-Model – Test levels (1/2)



Who's who?

Acceptance Test

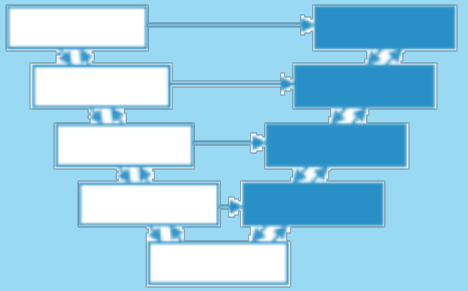
System Test

Integration Test

Component Test

- "I check whether the system as a whole fulfills the specified requirements."
- "I am there to check whether each software or hardware component fulfills its specification."
- "I check whether the components interact as defined in the technical system design."
- "I check whether the system has the agreed performance characteristics from the customer's point of view."

V-Model – Test levels (2/2)

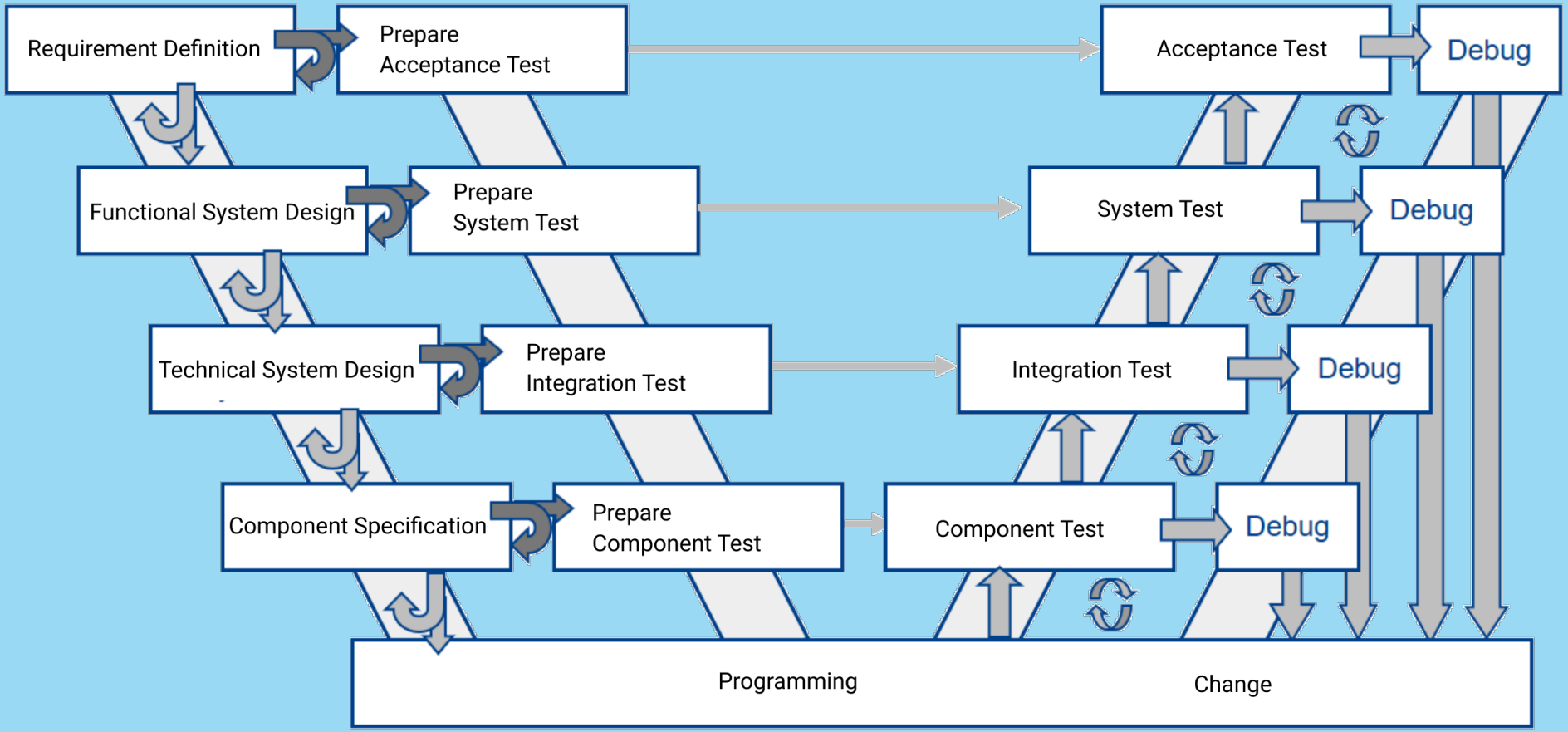


- **Component test**: Does every software or hardware component fulfill its specification?
- **(Component) Integration test***: Do components interact as specified in the technical system design?
- **System test**: Does the system as a whole fulfill the specified requirements?
- **Acceptance test**: Does the system exhibit agreed performance characteristics from the customer's point of view? inding errors is easiest on the same abstraction level

Additional type of integration test:

- **System integration test**: after system test, tests the interaction of different systems or between hardware and software
- **Unit Integration Test** = Component Integration Test

W-Modell – What is the reasoning behind it?



after Spillner/Roßner/Winter/Linz
Praxiswissen Softwaretest - Testmanagement
dpunkt, 2014, Chapter 3.3.2

Review, Previews, Documents

Test cases, Test environments

test, debug, change, re-test

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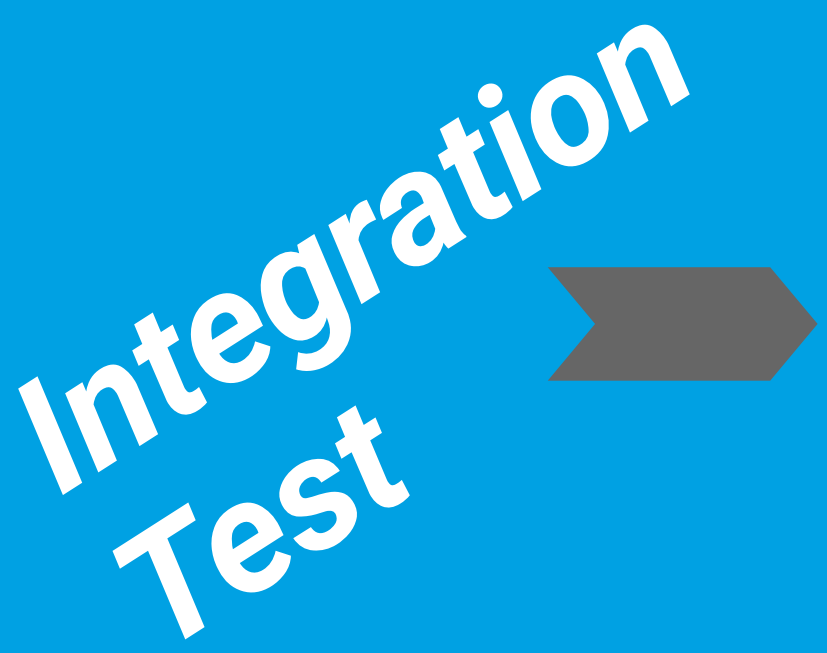
Chapter 2: Testing Throughout the Software Development Lifecycle

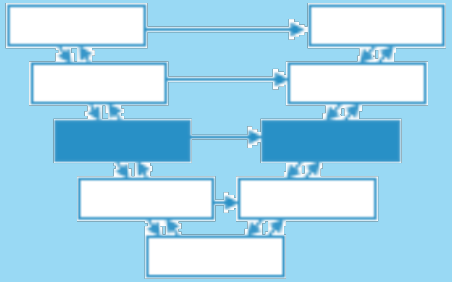


Testing in the Context of a Software Development Lifecycle

Test Levels and Test Types

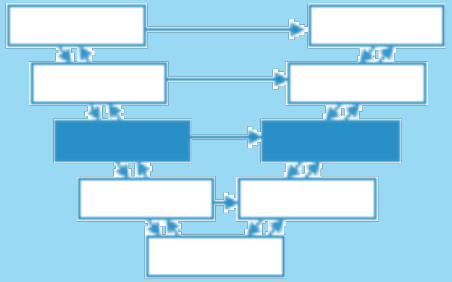
Maintenance Testing

Integration Test – Terms



- Integration test: second test stage after component testing
- Prerequisites
  - Component tests have concluded
  - Uncovered defects have been fixed
- Integration
  - Combine several components into larger subsystems
  - **Responsible**: developers, testers or **special integration teams**
- Integration test
  - Test whether the individual components work together
- Goals
  - Find interface failures
  - Finding failures in component interaction

Integration Test – Test Basis



- Software and system design
- Sequence diagrams
- Specifications of interfaces and communication protocols
- Use cases
- Architecture at component or system level
- Workflows
- External interface definitions (API)

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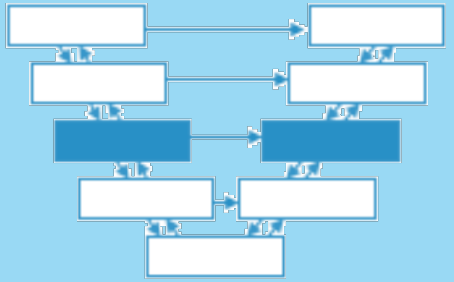
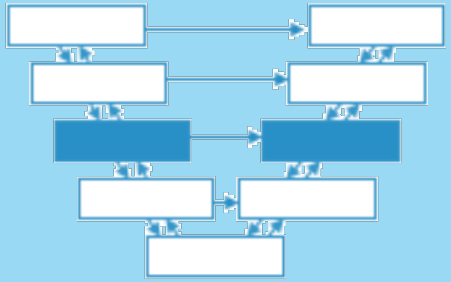
Integration Test – Test Objects



- Integrate & test individual modules step by step to form larger units
  - subsystems
  - databases
  - infrastructure
  - interfaces
  - APIs
  - Microservices
- Each subsystem can be the basis for the integration of larger units

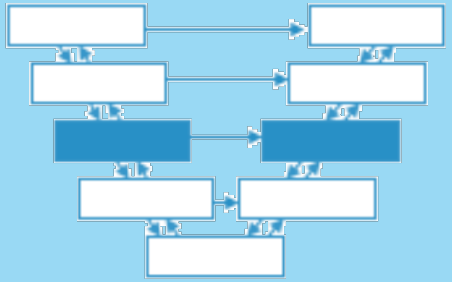
Integration Test – System Integration Test



Several potential integration levels:

- **Component integration test**
  - after component test
  - Focus: **Interaction of components**
  - Usually part of continuous integration
  - Often responsibility of the developers
- System integration test
  - After system test (also possible in parallel to system test)
  - Focus: **interaction of systems** (incl. hardware), packages, microservices
  - Interactions and interfaces of third parties can also be covered
  - **Often responsibility of the testers**

Integration Test – Defects and Failures



- Typical defects / failures:
  - Component transmits **incorrect data**, receiving component crashes (functional error of a component, incompatible interface formats, protocol error)
  - Receiving components interpret data incorrectly (functional error, contradictory or misinterpreted specifications)
  - Data is transferred correctly, but at the wrong time (timing problem) or at high frequency (throughput or load problem).
- None of these failures can be detected in the component test
- Failure only through interaction of components

Question

Why don’t we just skip component testing and do everything via integration test?

(this actually happens a lot in practice …)

What are the drawbacks of doing away with integration testing?

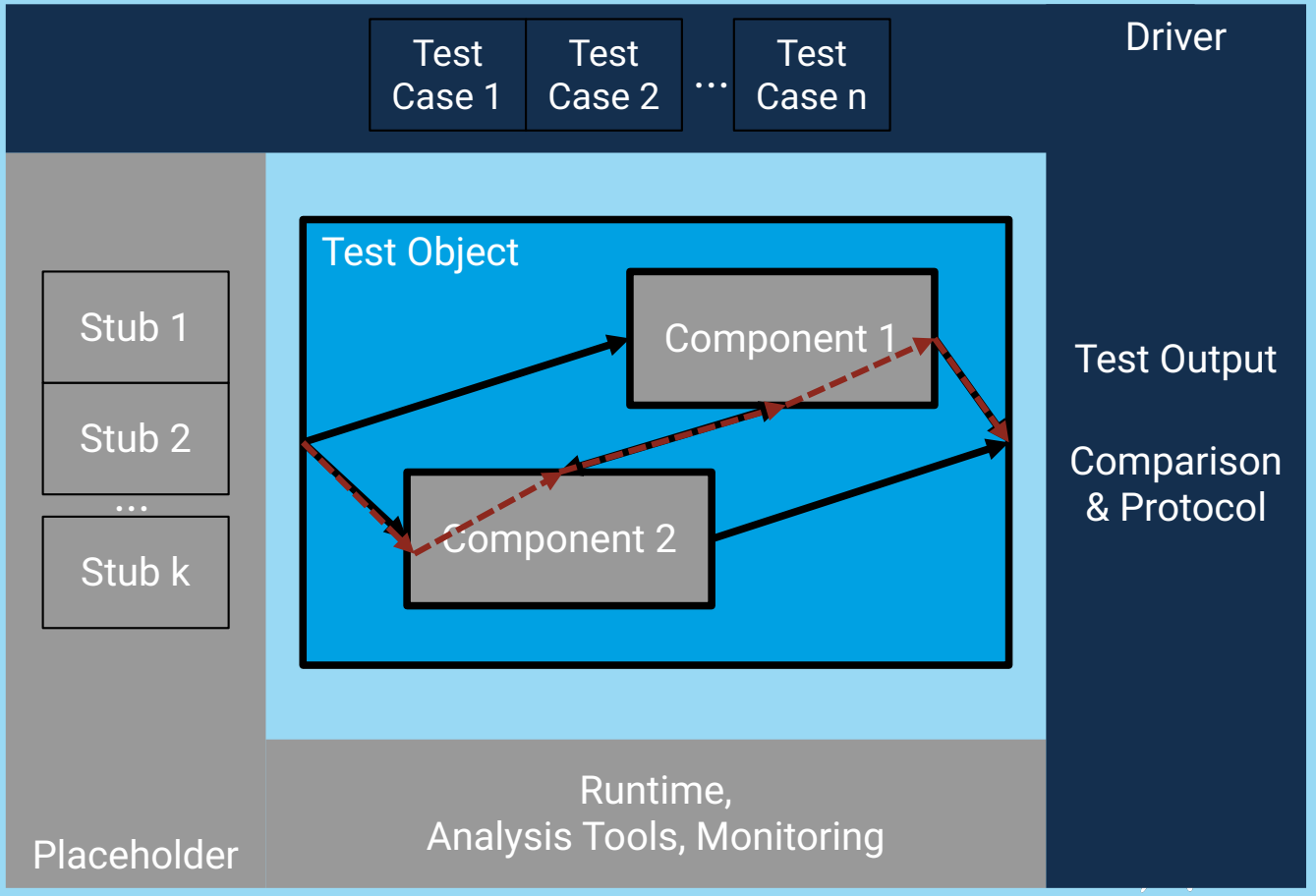
The greater the scope of an integration, the …

more difficult the isolation of defects

greater the time required to fix errors

Integration Test – Test Environment

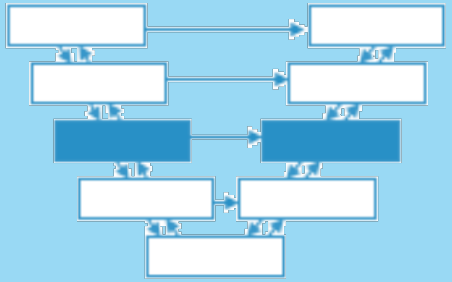
- Integration test with drivers: import test data, log results
- Reuse of existing component test drivers
- Additional diagnostic tool (monitors) for interface monitoring



Question

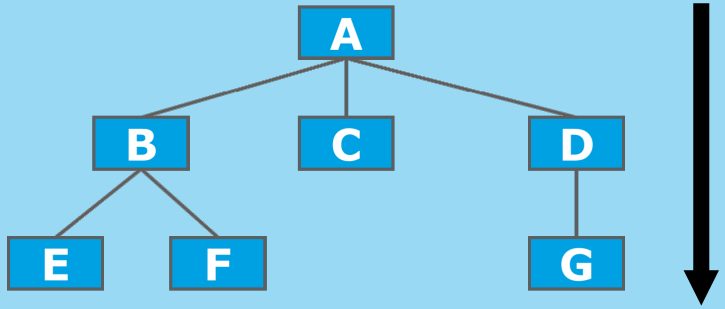
In what order do we integrate components?

Integration Test – Integration Test Strategy (1/5)



Top-down integration

- Test starts with component that calls others but is not called itself
- Successive integration of components in lower system layers
- Subordinate components replaced by placeholders
- Higher layer components under test used as driver



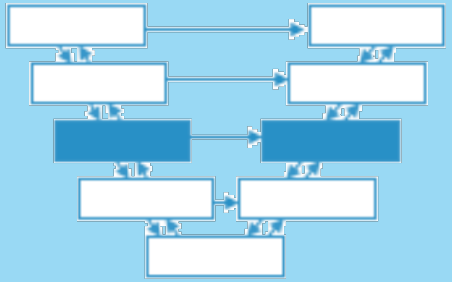
Advantage

- no / simple drivers required, as they consist of already tested components

Disadvantage

- subordinate, not yet integrated components must be replaced by placeholders

Integration Test – Integration Test Strategy (2/5)



Bottom-up integration

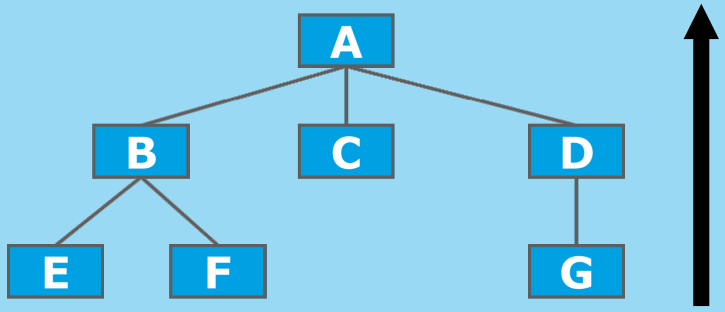
- Test starts with basic components that do not call others
- Successive integration of components with subsequent testing

Advantage

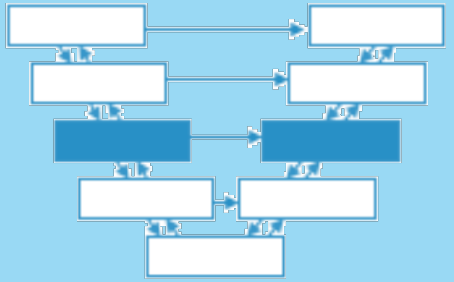
- No placeholders required.

Disadvantage

- Superordinate components must be simulated by drivers

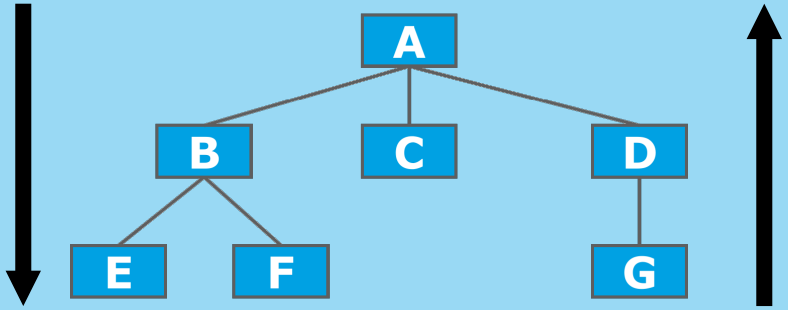


Integration Test – Integration Test Strategy (3/5)

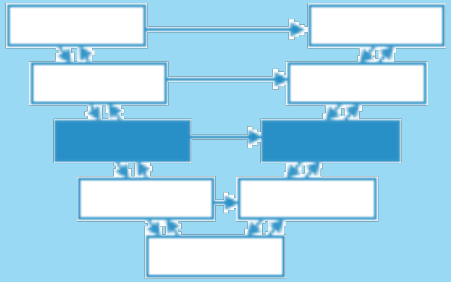


Comments

- Top-down or bottom-up approach can only be used for hierarchically structured systems (not super common)
- In practice, a mix of both integration strategies is commonly used



Integration Test – Integration Test Strategy (4/5)



Ad-hoc Integration

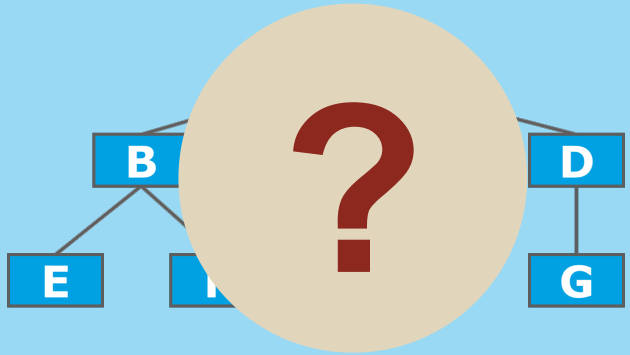
- Integrate components in (random) order of completion
- Check feasibility of integration directly after component test

Advantage

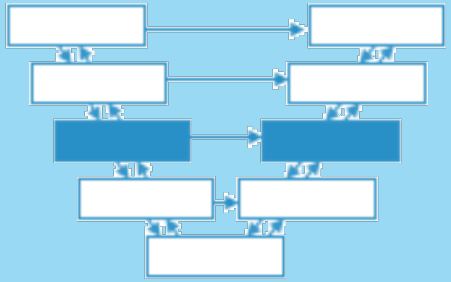
- Time saving, as earliest possible integration

Disadvantage

- Placeholders and drivers required

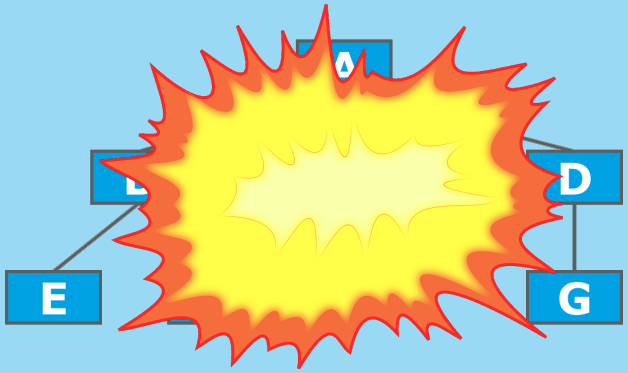


Integration Test – Integration Test Strategy (5/5)



Non-incremental Integration – *big-bang* Integration

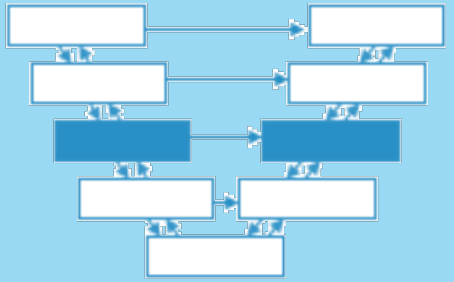
- Wait until all software components have been developed
- Integrate everything at once. At worst, without component tests
- Extreme version: integrate software and hardware elements at once

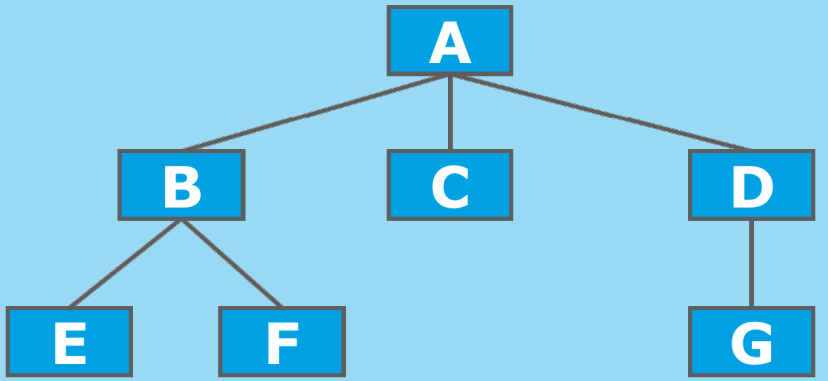


Disadvantages

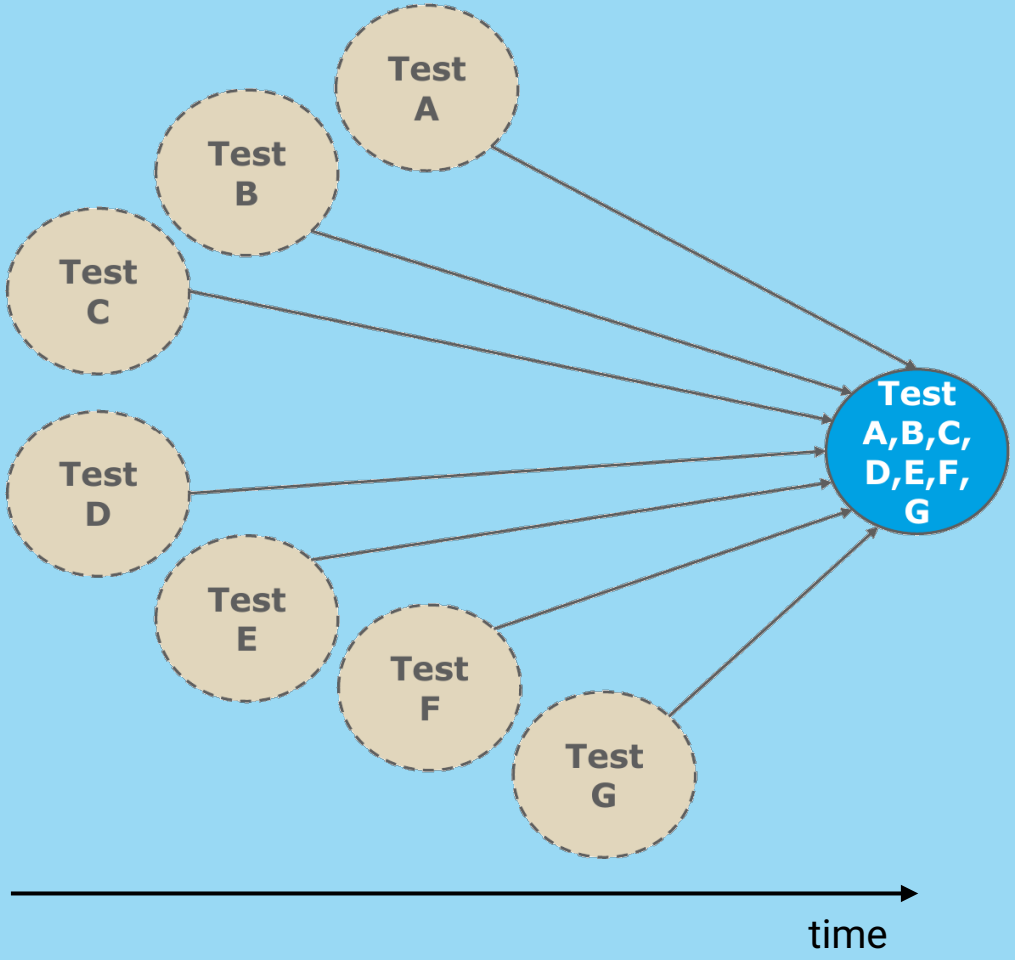
- Waiting time until the *big bang* is lost testing time
- Failures occur all at once
- Very difficult to ensure that the system works at all
- Finding and correcting defects is extremely difficult

Integration Test – Integration Test Strategy with Example (1/4)

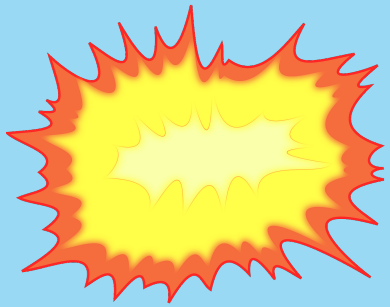


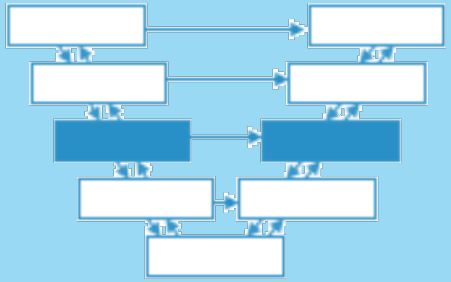


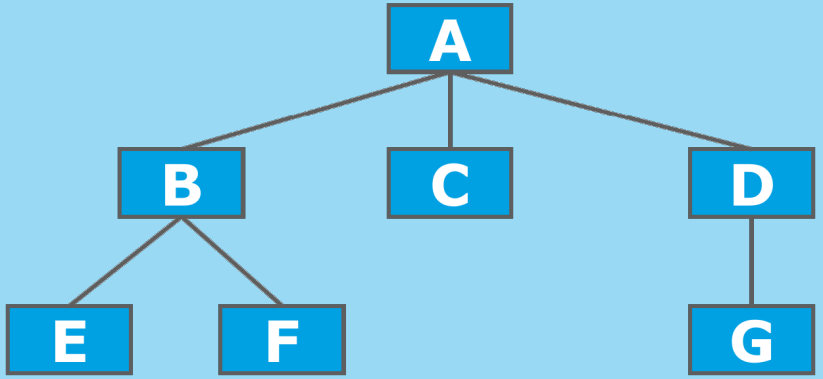
Integration Test – Integration Test Strategy with Example (2/4)



***big-bang* Integration**







Component Test

Integration Test

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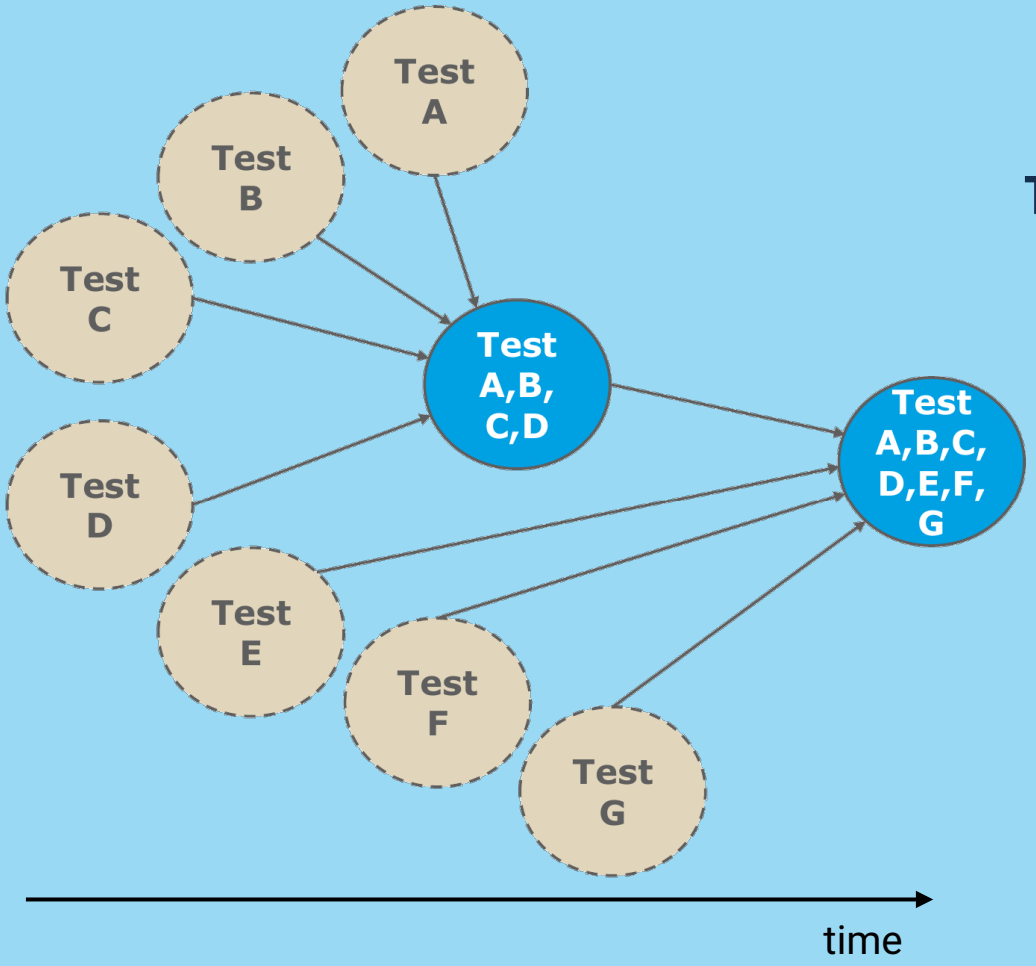
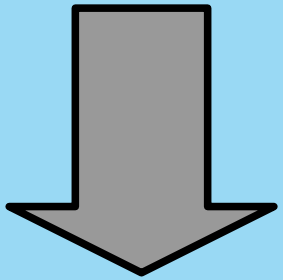
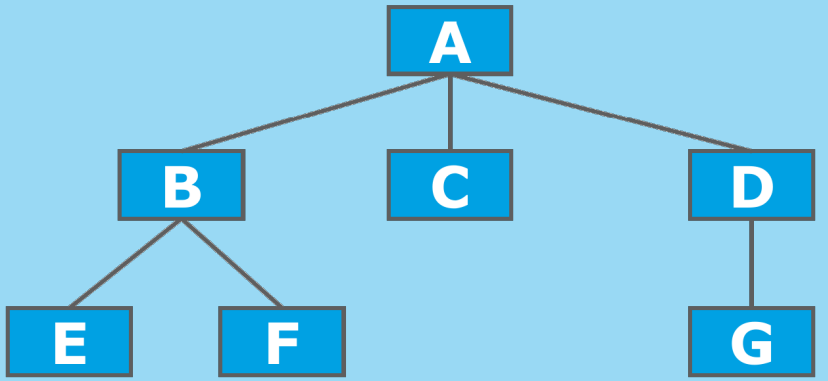
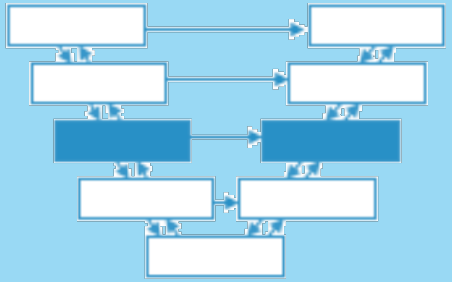
Integration Test – Integration Test Strategy with Example (3/4)



Top-Down Integration







Component Test

Integration Test

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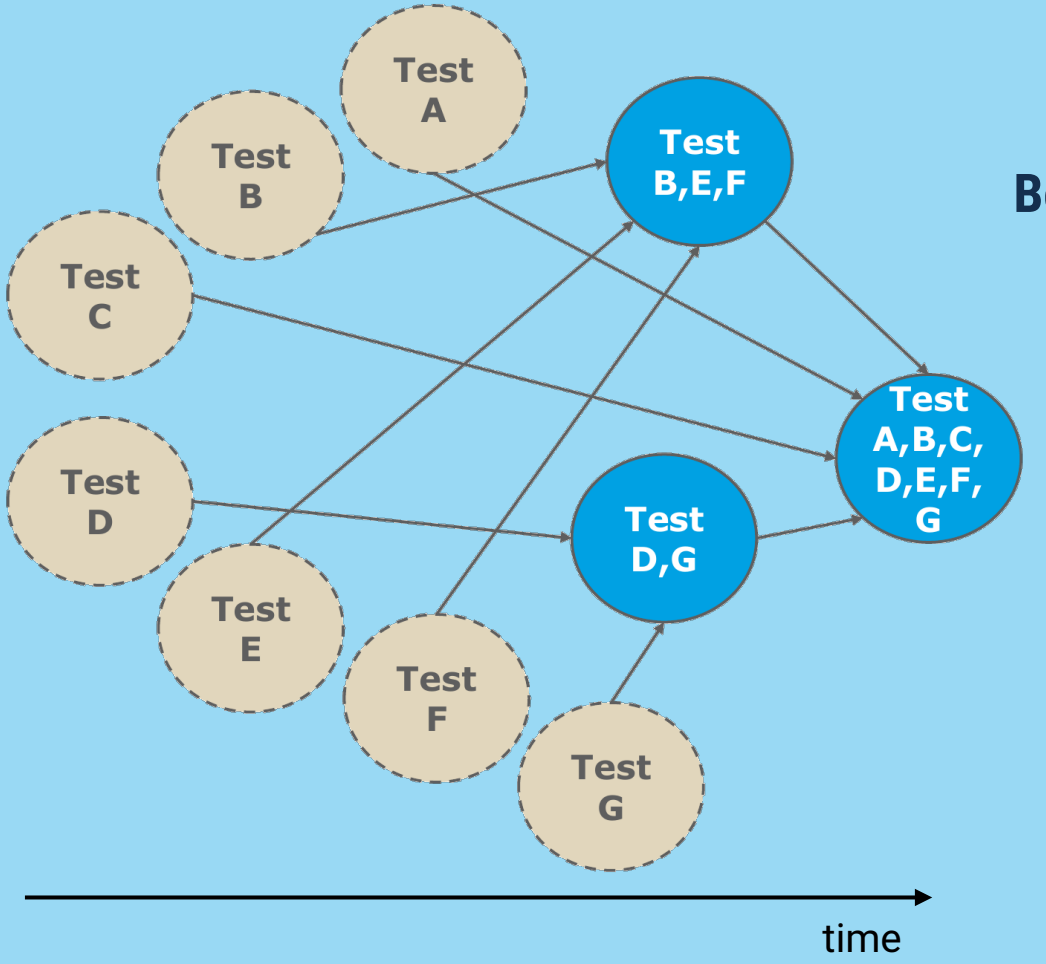
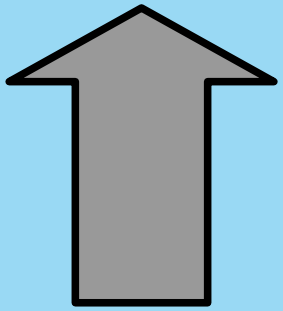
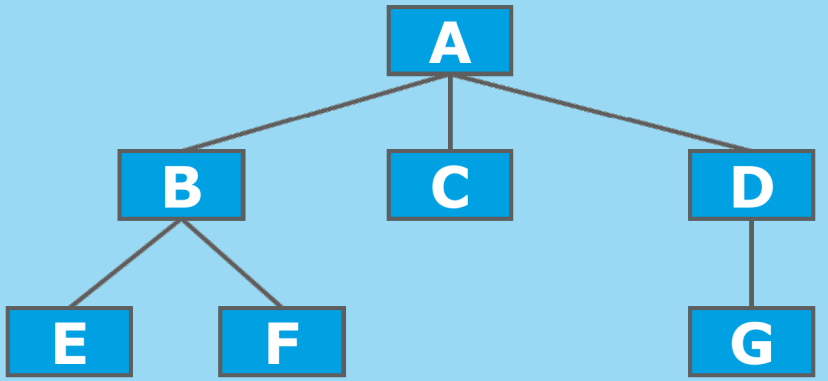
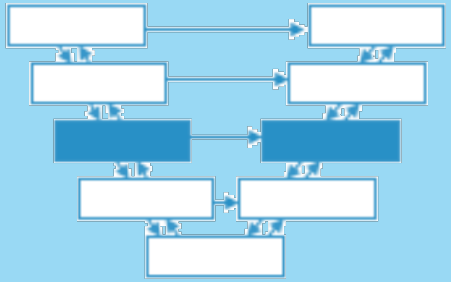
Integration Test – Integration Test Strategy with Example (4/4)



Bottom-Up Integration







Component Test

Integration Test

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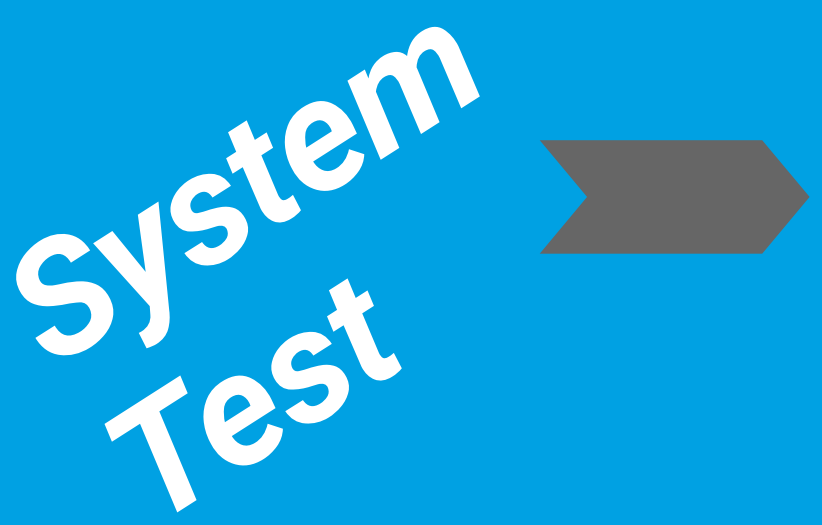
Chapter 2: Testing Throughout the Software Development Lifecycle



Testing in the Context of a Software Development Lifecycle

Test Levels and Test Types

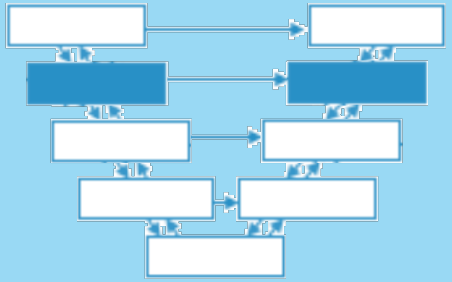
Maintenance Testing

Question

Now that we have tested all components by themselves and integrated…

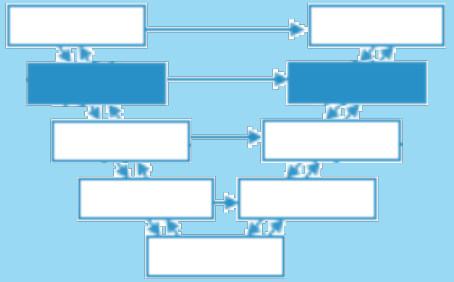
Why do we need to run a system test?

System Test – Terms



- System test: after the completion of integration test
- Focus: does the product fulfill the specified requirements?
- Motivation
  - Previous test stages: Testing from the perspective of the software manufacturer
  - System test: Testing from the perspective of customers and users
  - Requirements fully and appropriately implemented?
  - System functions and properties can often only be tested after integration of **all (!) system** components

System Test – Test Objectives



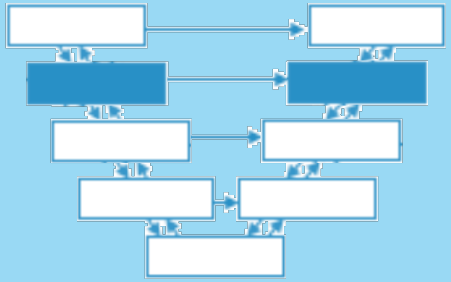
- **Consideration of system as a whole (functional and non-functional)**
- Risk reduction, finding of failures (and defects)
- On failure: Do not pass on to higher test levels / production
- Partially: Verification of data quality
- Create confidence in the quality of the system as a whole
- Collect information for release decision

Question

What can be used as a test basis for the system test?

System Test – Test Basis

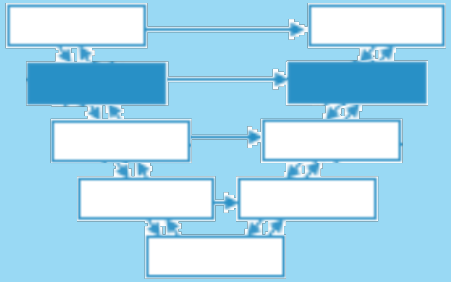
Examples for test basis in system testing



- (non-)functional requirement specifications
- Use cases, epics and user stories
- Risk analysis reports
- System behavior Models
- Documentation and user guides

System Test – Test Objects

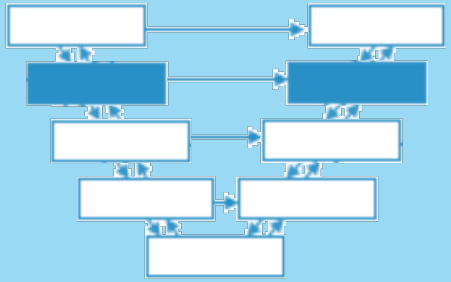
Common test objects in system testing



- Systems under test (SUT)
- Applications
- Hardware/software systems
- Operating systems
- System configuration and configuration data

System Test – Common Defects and Failures

Common failures and defects in system testing

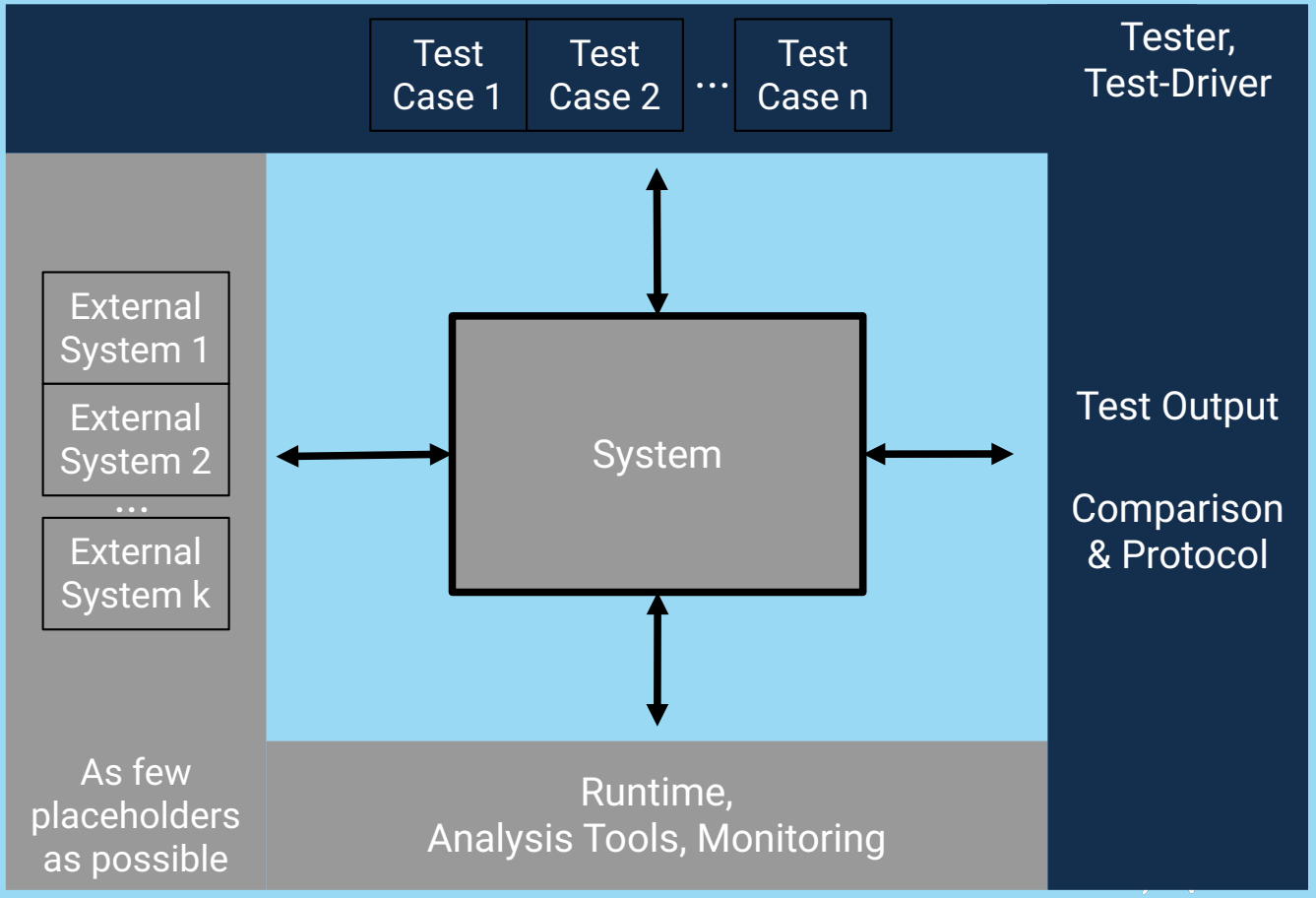


- Failure to correctly complete execution of functional tasks
  end-to-end tasks
- Failure of the system to work properly in the system environment(s)
  System environment(s)
- **System does not work as described in documentation**

- e.g. incorrect calculation results,
  incorrect / unexpected (non-)functional system behavior,
  incorrect control and/or data flows within the system

System Test – Test Environment

- Test environment should closely resemble later production environment
- Wherever possible: Use hardware or software that will actually be used



Question

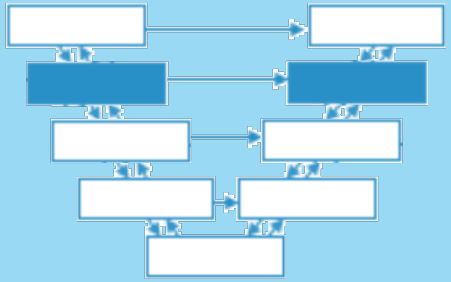
Should system tests be run in production?

Why? Why not?

Software maturing at the customer’s.

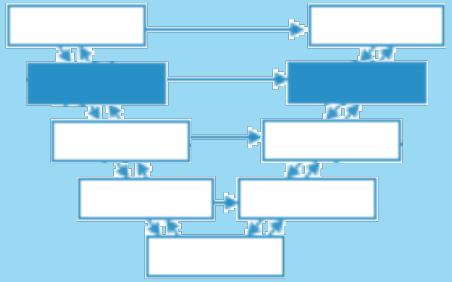
System Test – Test Environment

Disadvantages when testing in production



- Failures can affect the customer's productive environment
- Possible consequence: expensive system failures and data loss in customer system
- No / little control over configuration of the production environment; meaning potential changes to test conditions due to operation running parallel to the test
- System tests are difficult to reproduce / no longer reproducible

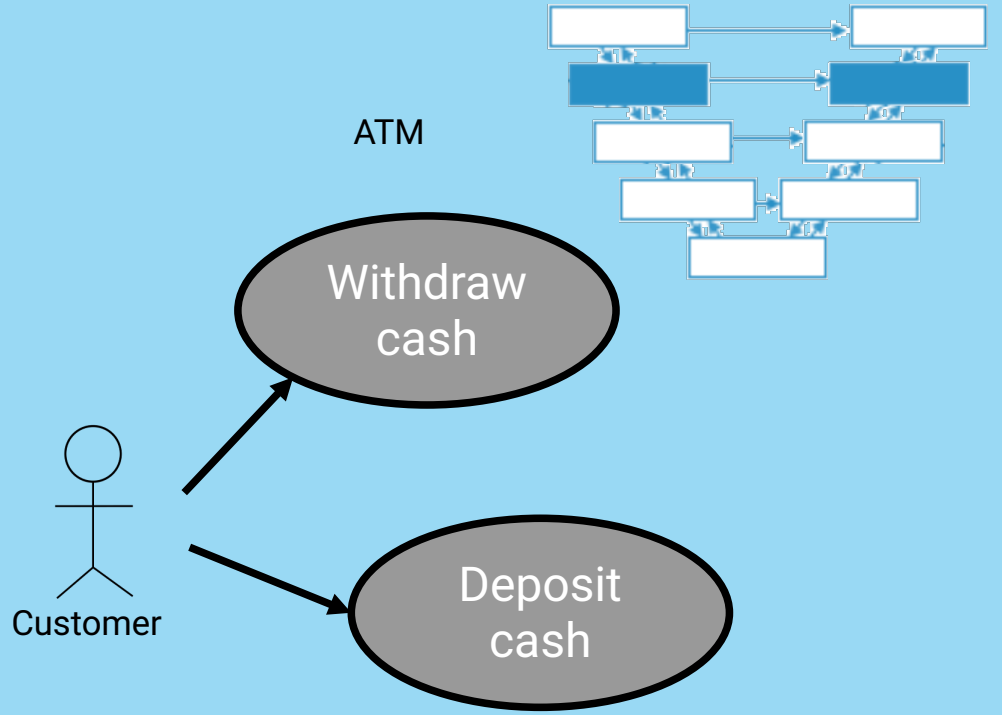
System Test – Test Strategy (1/2)



- Requirement-based testing
  - Test basis: Requirements document
  - At least one system test case derived and documented per requirement
  - Usually more than one test case to test a requirement
- Business process-based testing
  - **Create test scenarios that emulate typical business processes**
  - Business process analysis shows the **relevant business processes** (incl. context, people, companies, external systems, ...)
  - Priority based on frequency / relevance of business processes

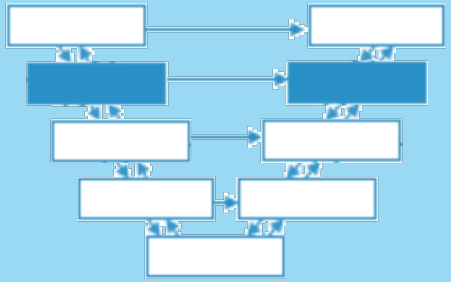
System Test – Test Strategy (2/2)

- Use case based testing
- System test cases represent typical system use
- User profile per user group with common usage pattern
- Derive test scenarios from usage patterns
- Priority of test scenarios based on frequency of actions monitored in operation



System Test – Test Strategy for NFRs (1/3)

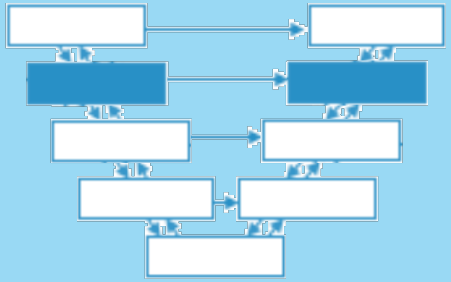
Non-functional requirements



- Performance test
  - Measurement of response time for use cases (with increasing load)
- Load test
  - Behavior of a system under changing load - usually between low load, typical load and peak load
- Volume/mass test
  - Check system behavior in relation to data volume (e.g. for very large files)

System Test – Test Strategy for NFRs (2/3)

Non-functional requirements



- Test of (data) security
  - against unauthorized system access or data access
- Reliability test
  - Continuous operation
    (failures per operating hour with usage profile x, ...)
- Resilience and Robustness test
  - against incorrect operation, incorrect programming, hardware failure
  - Testing of error handling and restart behavior
  - → ***Chaos Testing***

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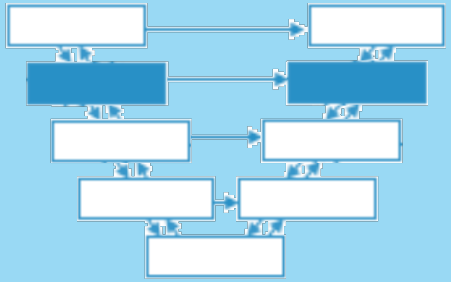
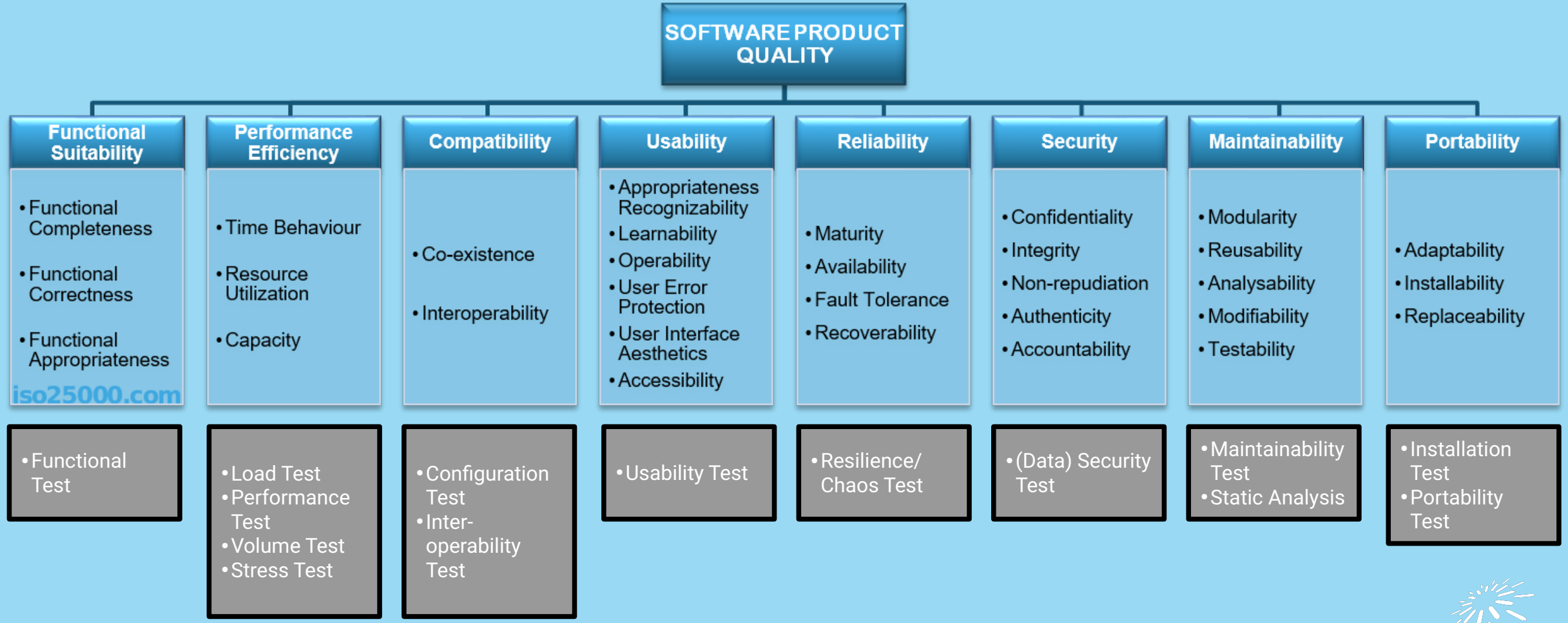
System Test – Test Strategy for NFRs (3/3)

Non-functional requirements

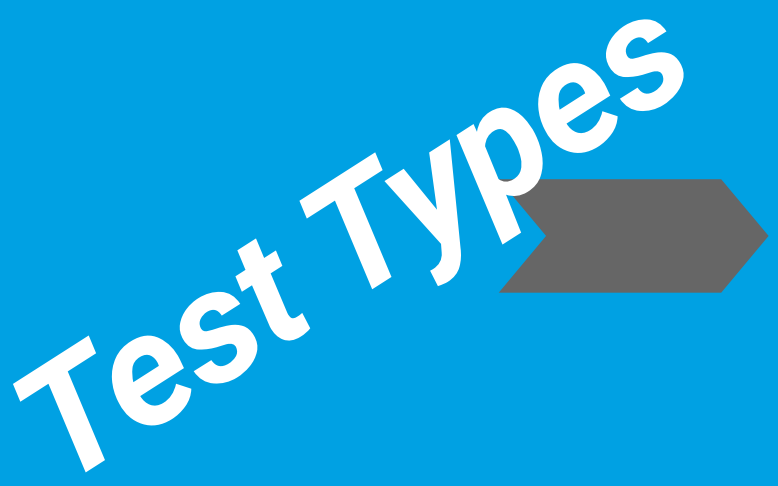


- Compatibility test / data conversion test
  - Check compatibility with existing systems
  - Import/export of databases
- Configuration test
  - Focus: different configurations of the system
- Usability test / usability test
  - Testing usability, comprehensibility of system output, ...
- Checking documentation
  - Compliance with system behavior (e.g. operating instructions)
- Check for changeability/maintainability
  - Comprehensibility of development documents, system structure, etc.

System Test – Quality Attributes according to ISO 25010



Chapter 2: Testing Throughout the Software Development Lifecycle



Testing in the Context of a Software Development Lifecycle

Test Levels and Test Types

Maintenance Testing

Learning Objectives for Testing Throughout the Software Development Lifecycle

(according to Certified Tester Foundation Level Syllabus, English Version V4.0)

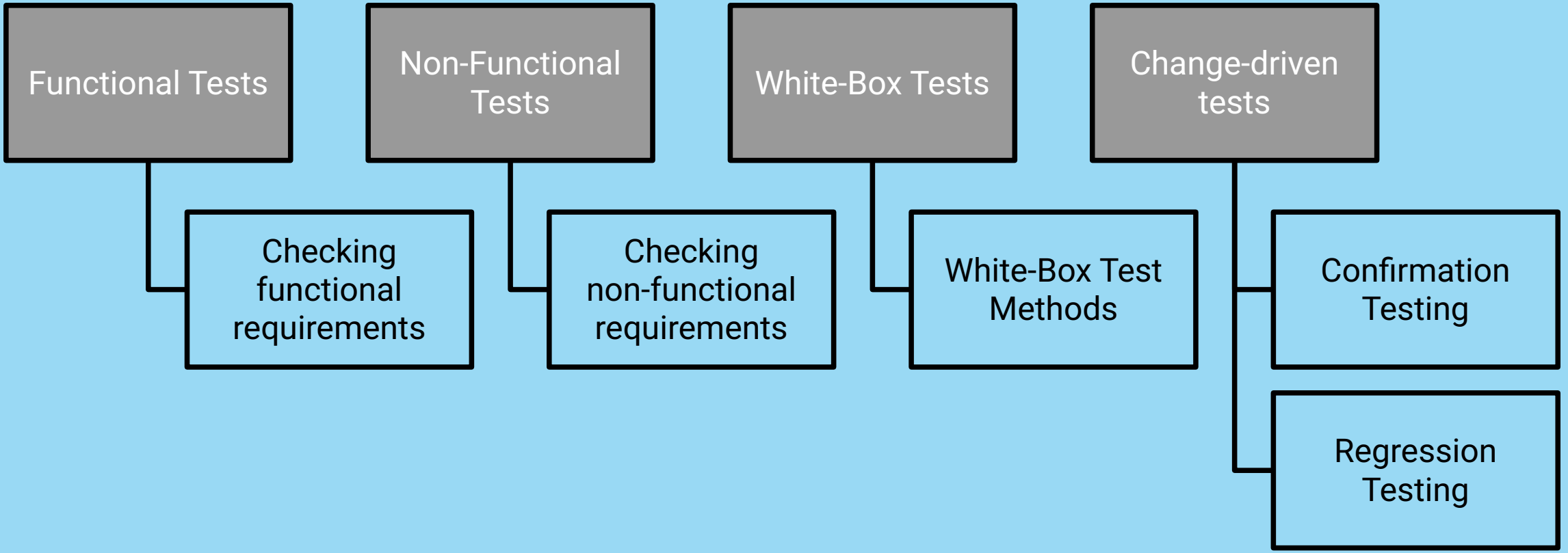
- **2.2 Test Levels and Test Types**
- FL-2.2.1 (K2) Distinguish the different test levels
- FL-2.2.2 (K2) Distinguish the different test types
- FL-2.2.3 (K2) Distinguish confirmation testing from regression testing
- **2.3 Maintenance Testing**
- FL-2.3.1 (K2) Summarize maintenance testing and its triggers

Test Types – Terms (1/2)

Test types group test activities according to test objectives:

- **Functional quality attributes**: complete, correct, appropriate?
- **Non-functional quality attributes**: reliable, performant, usable?
- Correctness and completeness of the **component** architecture
- Effects of **changes**

Test Types – Terms (2/2)



Test Levels and Test Types

- Each test type possible at each test level
  - Component test
  - Integration test
  - System test
  - Acceptance test
- Intensity of specific test types may vary per test level

Question

What is the difference between a Confirmation Test and a Regression Test?

Confirmation Test

- **Goal: Confirmation that defect has been removed**

- Defect fixed → perform tests that failed due to defect!
- New tests possible if defect was the absence of some functionality

Regression Test (1/3)

- Retesting of already tested software
- **Goal**: no new / previously masked defects in **unchanged areas**

- **Regression testing does NOT have the goal of finding errors!**

- Consider using regression tests even if software environment has changed
- Scope of the regression test?

Question

What is a good scope for regression tests?

Question

Impact Analysis

→ simple local changes may have unexpected effects

→ side effects on any (even distant) software possible

What is a good scope for regression tests?

Regression Test (2/3)

- Complete regression test
  - Repetition of all existing test cases
  - Same significance as test on initial version of the software
  - Also necessary if system environment has changed

- In practice: complete regression test almost always too expensive

Regression Test (3/3)

- Selection of regression test cases
  - Repeat high priority tests only
  - Omit special cases in functional tests
  - Focus on specific configurations (e.g. language, OS)
  - Focus on software impacted by changes

- Regression test suites often executed with few changes
- Excellent candidates for test automation

Perform
confirmation tests and regression tests
(repeatedly) at all test levels!

Chapter 2: Testing Throughout the Software Development Lifecycle

Testing in the Context of a Software Development Lifecycle

Test Levels and Test Types

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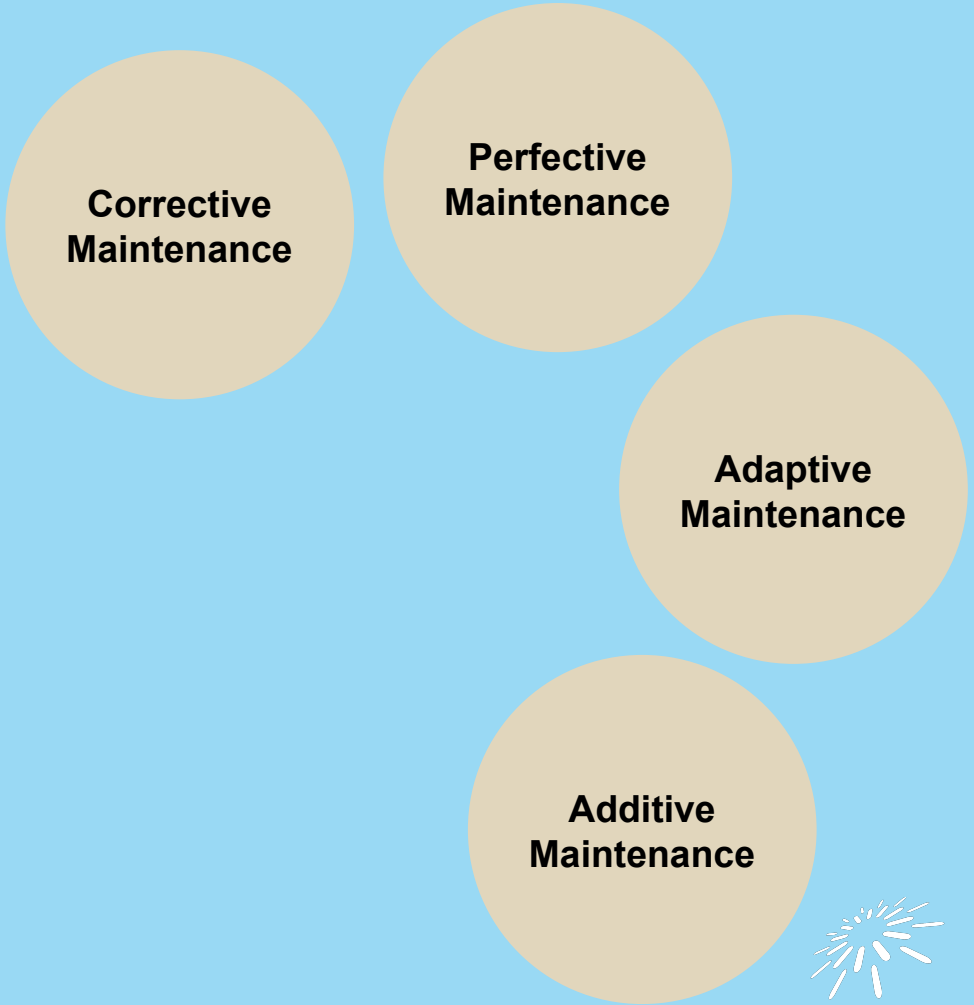
Testing a new Product Version (Maintenance Test)

- End of development after passing acceptance test and delivery?

- Reality:
  - Initial delivery is only the beginning of the life cycle
  - Often in use for years / decades after installation
  - Multiple changes to software, environment or configuration during this time
  - New version to be tested each time

Some Reasons for Maintenance – Which is which?

- Planned ongoing development
- Performance improvement
- Changing the environment
- Troubleshooting
- System refactoring
- Changes due to other changes in a neighboring system
- Implementation of a planned but unfinished functionality
- Extensions for a planned market expansion



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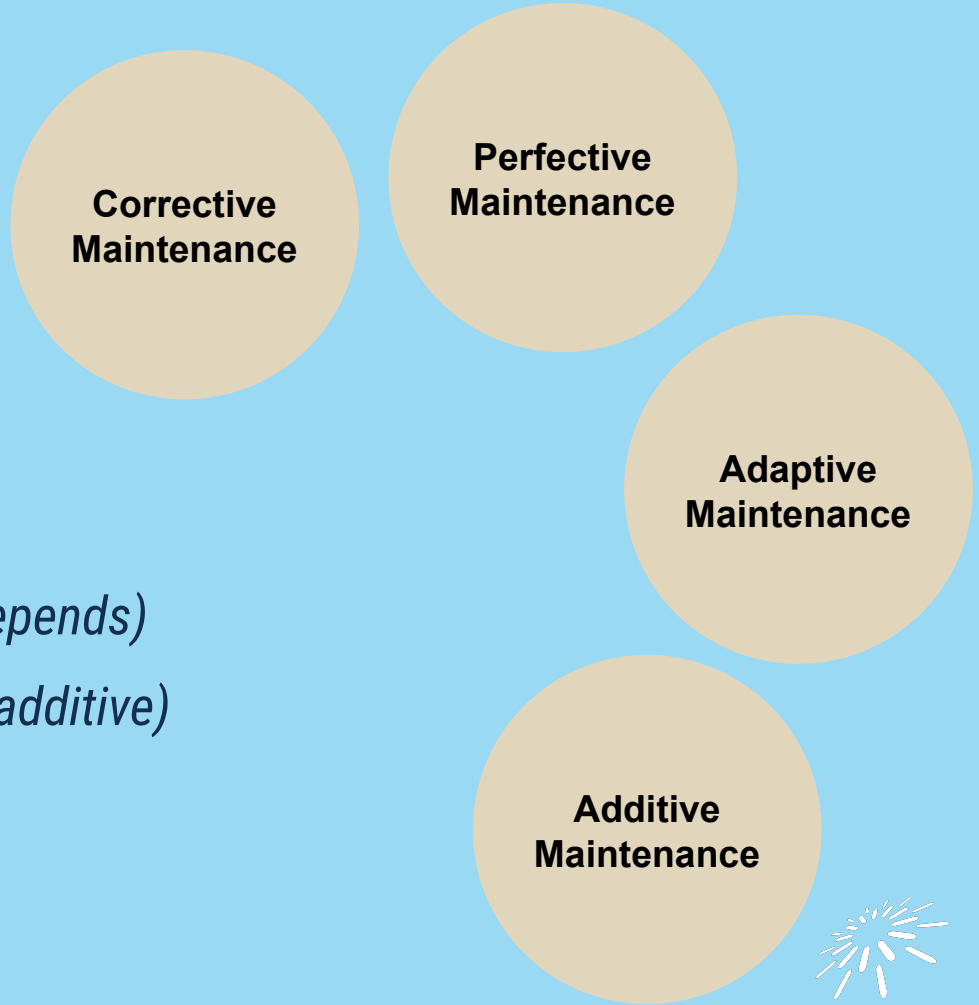
Some Reasons for Maintenance – Which is which?

- Planned ongoing development *(additive)*
- Performance improvement *(perfective)*
- Changing the environment *(adaptive)*
- Troubleshooting *(corrective)*
- System refactoring *(perfective)*
- Changes due to other changes in a neighboring system *(depends)*
- Implementation of a planned but unfinished functionality *(additive)*
- Extensions for a planned market expansion *(adaptive)*



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Reasons for Maintenance (1/2)

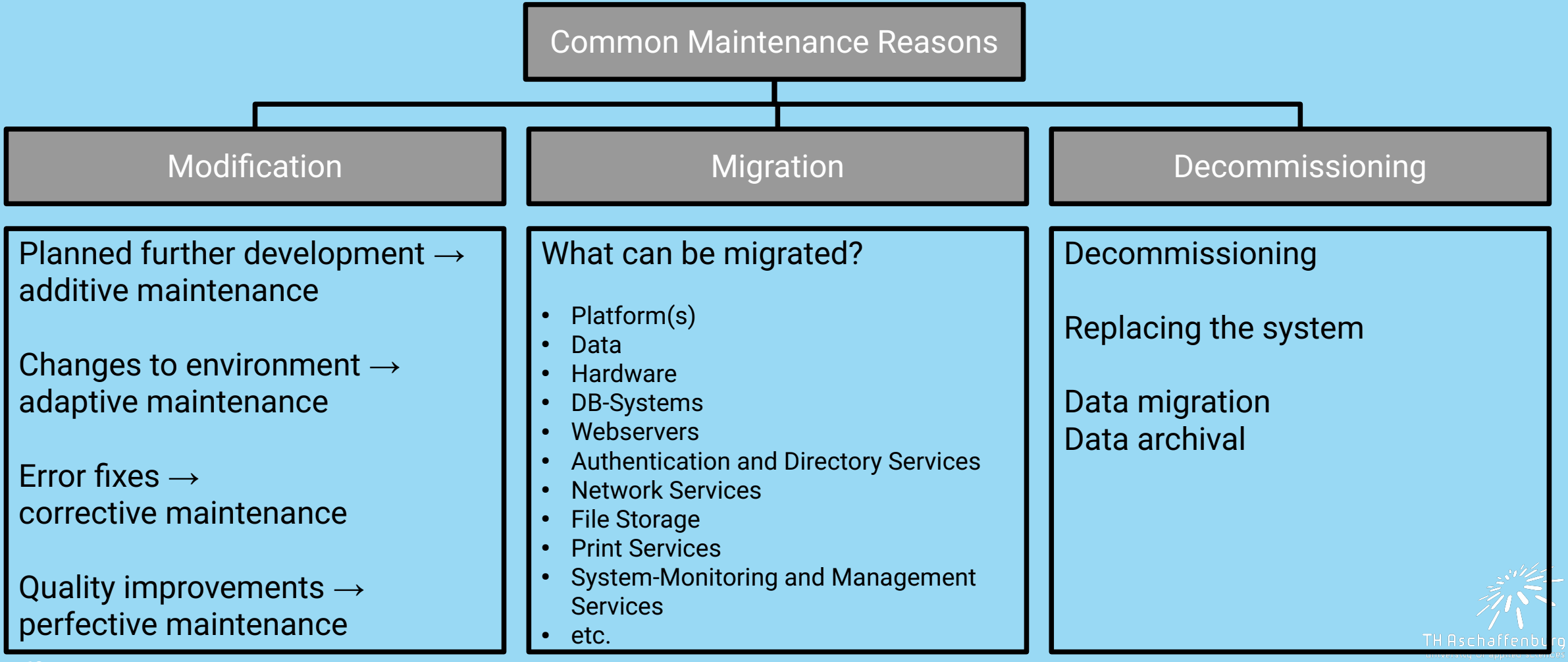
- Planned ongoing development *(additive maintenance)*
  - Modification and extension work planned from the outset
  - Normal product development
  - Software project not completed with delivery of the first version

- Troubleshooting *(corrective maintenance)*
  - Errors that occurred operation
  - Emergency fixes (“hot fixes”)

Reasons for Maintenance (2/2)

- Changing the environment *(adaptive maintenance)*
  - OS updates
  - DB management system updates
  - Upgrades of commercial software
  - Patches of external components
  - etc.
- Improving quality *(perfective maintenance)*
  - Improvement of quality factors, e.g. maintainability, performance, usability without changes in functional scope

Software Maintenance



Maintenance Test during Modification

- Maintenance release may require maintenance tests in several test stages
- Scope of maintenance tests depends on:
  - Change risk level (e.g. communication intensity)
  - Existing system size
  - Size of change
- Use impact analysis and traceability!

Maintenance Test during Migration/Decommissioning

- Maintenance test during migration
  - Includes tests during operation of new environment / changed software
- Maintenance test during decommissioning for ...
  - Data migration (to new system): Example migration of customer data in different formats
  - Archiving (for long retention periods)
  - Recovery procedure after archiving for long retention periods
  - Regression tests for functionality that remains in operation
- Conversion tests necessary
  - Test of data conversion software
  - Test of converted data

Take Away – Chapter 2 (1/2)

- General V-model
  - Test levels: Component test, integration test, system test (and acceptance test)
  - differentiates between verifying and validating tests
- **Component tests** test individual software modules
- **Integration tests** check the interaction of tested software modules
- Functional and non-functional **system tests** consider entire system
- **Acceptance test**: Client checks product for user acceptance and against acceptance criteria

Take Away – Chapter 2 (2/2)

- Basic test types
  - functional and non-functional test
  - structural and change-related testing
- Modification of software during life cycle
- **Testing of every changed version necessary!**
- Determine the scope of regression tests through impact analysis and risk assessment!

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Until next time!

Collaboration, Quality, Test

Prof. Dr. Jóakim von Kistowski

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